LMT-GP: Combined Latent Mean-Teacher and Gaussian Process for Semi-supervised Low-light Image Enhancement

Supplementary Material

In this document, we first offer more details of the network architecture. Then we present ablation studies to evaluate the effectiveness of different kernel functions. Furthermore, we show LLIE experimental results on three unlabeled datasets LIME-Data [1], DICM [2], and MEF [3], to further validate the generalization ability of our method. Finally, we provide additional results on LOL, VE-LOL, Darkface, and ACDC datasets to justify our strengths.

1 Network Architecture

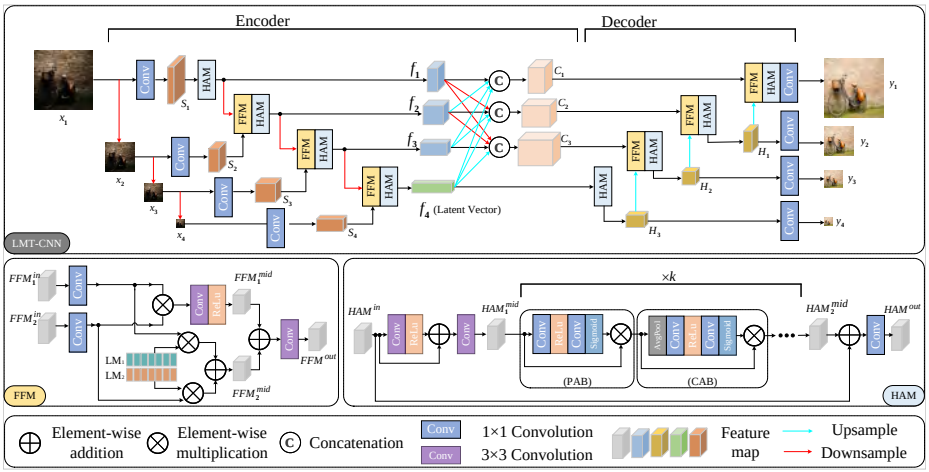


Fig. 1: Illustration of LMT-CNN.

In this section, we introduce the details of the network architecture in Section 3.3 of the main paper. As shown in Fig. 1, the student and teacher networks share the same LMT-CNN architecture. LMT-CNN consists of an encoder and a decoder with four-layer pyramid structures.

We use the encoder to obtain intermediate features. Specifically, we first downsample the input image x_1 to obtain images of different scales, denoted as x_2, x_3, and x_4. Then we use a 1×1 convolution to extract S_i $(i \in \{1, 2, 3, 4\})$ from x_i $(i \in \{1, 2, 3, 4\})$, respectively. The intermediate feature f_i $(i \in \{1, 2, 3, 4\})$ can

be defined as follows:

$$f_i = \begin{cases} HAM_i^{En}(S_i), & i = 1 \\ HAM_i^{En}(FFM_i^{En}(S_i, (f_{i-1})^{\downarrow}), & i \in \{2,3,4\} \end{cases}, \tag{1}$$

where $HAM_i^{En}(\cdot)$ and $FFM_i^{En}(\cdot)$ represent the outputs of HAM and FFM in the *i-th* layer of the encoder, respectively; $(\cdot)^{\downarrow}$ represents the downsampling operation; f_4 is the latent vector. We perform different upsampling or downsampling operations on f_1, f_2, f_3, and f_4. Then we apply a concatenation operation to obtain C_i ($i \in \{1,2,3\}$) for the decoder. C_i can be denoted as follows:

$$C_i = \begin{cases} Concat(f_1, (f_2)^{\uparrow}, (f_3)^{\uparrow}, (f_4)^{\uparrow}), & i = 1 \\ Concat((f_1)^{\downarrow}, f_2, (f_3)^{\uparrow}, (f_4)^{\uparrow}), & i = 2 \\ Concat((f_1)^{\downarrow}, (f_2)^{\downarrow}, f_3, (f_4)^{\uparrow}), & i = 3 \end{cases}, \tag{2}$$

where $(\cdot)^{\uparrow}$ and $Concat(\cdot)$ represent upsampling and concatenation operations, respecitvely.

We use the decoder to obtain the multi-scale output image y_i ($i \in \{1,2,3,4\}$). It can be defined as follows: 0.8

$$y_i = \begin{cases} Conv^{1\times1}(HAM_i^{De}(FFM_i^{De}(C_i, (H_i)^{\uparrow}))), & i = 1 \\ Conv^{1\times1}(H_{i-1}), & i \in \{2,3,4\} \end{cases}, \tag{3}$$

$$H_i = \begin{cases} HAM_{i+1}^{De}(FFM_{i+1}^{De}(C_{i+1}, (H_{i+1})^{\uparrow})), & i \in \{1,2\} \\ HAM_{i+1}^{De}(f_{i+1}), & i = 3 \end{cases}, \tag{4}$$

where $HAM_i^{De}(\cdot)$ and $FFM_i^{De}(\cdot)$ represent the outputs of HAM and FFM in the *i-th* layer of the decoder, respectively; $Conv^{1\times1}(\cdot)$ represents 1×1 convolution operation.

FFM. We use FFM to fuse feature maps from different scales. We first employ a 1×1 convolution to individually extract features from the input feature maps, FFM_1^{in} and FFM_2^{in}. Then, we perform two parallel operations: 1) perform element-wise multiplication between the two extracted features, followed by a 3×3 convolution operation and a ReLU activation function to obtain the feature map FFM_1^{mid}; 2) each extracted feature undergoes an element-wise multiplication with a learnable matrix (LM) that has the same dimension as the feature itself, and the results undergo element-wise addition to produce the feature map FFM_2^{mid}. Finally, we perform the element-wise addition and a 3×3 convolution to obtain the output FFM^{out}.

HAM. We use HAM to extract spatial and channel features. We first extract features by performing a residual structure with a 1×1 convolution and a ReLU activation function. Then the extracted features undergo another 1×1 convolution to obtain HAM_1^{mid}. HAM_1^{mid} can be defined as follows:

$$HMA_1^{mid} = Conv^{3\times3}(HAM^{in} + ReLU(Conv^{3\times3}(HAM^{in}))), \tag{5}$$

where $Conv^{3\times3}(\cdot)$ and $ReLU(\cdot)$ represent the 3×3 convolution operation and the ReLU activation function, repectively. Then, we utilize k times position attention blocks (PABs) and the channel attention blocks (CABs) to obtain HAM_2^{mid} which contains spatial and channel features. The outputs of PAB and CAB can be defined as follows: 0.9

$$PAB(z_1) = \sigma(Conv^{1\times1}(ReLU(Conv^{1\times1}(z_1)))) \otimes z_1, \tag{6}$$

$$CAB(z_2) = \sigma(Conv^{1\times1}(ReLU(Conv^{1\times1}(Avg(z_2))))) \otimes z_2, \tag{7}$$

where $Avg(\cdot)$ and $\sigma(\cdot)$ represent the average pooling operation and the sigmoid activation function, repectively; z_1 and z_2 represent the inputs of PAB and CAB. Finally, we perform the element-wise addition between HAM^{in} and HAM_2^{mid}, followed by a 1×1 convolution operation to obtain the output HAM^{out}.

2 Experiment

2.1 Evaluation of Kernel Functions

This section is an extension of the ablation study in Section 4.3 of the main paper. The kernel function we use to compute joint distribution is the squared exponential kernel (SE) as described in Formula (10) of the main paper. Here, we evaluate the performance of SE and compare with other kernel functions, such as linear kernel (LIN) and rational quadratic kernel (RQ). As shown in Table 1, the SE kernel function exhibits excellent performance.

Table 1: Quantitative results of different kernel functions.

	LIN	SE	RQ
PSNR ↑	23.39	**24.12**	23.31
SSIM ↑	**0.903**	0.902	0.902
LPIPS ↓	0.116	**0.112**	0.117

2.2 Evaluation on Unlabeled Low-light Datasets

To further demonstrate the generalization ability of our method, we conduct experiments on unlabeled low-light datasets LIME-Data [1], DICM [2], and MEF [3]. We use non-reference image quality assessment (IQA) metrics, namely NIQE [4], MANIQA [10], NIMA [5], and CLIPIQA [6], which evaluate the perceived quality of unlabeled low-light images based on on the human subjective perception. To evaluate the performance of LLIE, we select the top 5 methods (including ours) with the highest PSNR scores on the LOL dataset and then test them on these datasets.

Table 2 shows the quantitative results. Our method achieves the best NIMA and CLIPIQA scores on the LIME-Data and MEF datasets. We also obtain the best NIQE score on the DICM dataset. Furthermore, our method achieves the best MANIQA scores on the DICM and MEF datasets.

Fig. 2 shows the qualitative results. These samples are captured in different lighting environments, including incandescent lighting, indoor natural lighting,

Table 2: Quantitative results of experiments on the LIME-Data, DICM, and MEF datasets. The best result is in **red** whereas the second-best is in **blue**.

Dataset	Method	Metrics			
		NIQE↓	MANIQA↑	NIMA↑	CLIPIQA↑
LIME-Data	GLADNet [8]FG'18	5.398	0.428	3.302	0.243
	KinD [11] MM'19	5.485	0.465	3.152	0.325
	LLFormer [7] AAAI'23	**4.142**	0.587	4.074	0.436
	LLFlow [9]AAAI'22	4.892	**0.630**	**4.405**	**0.438**
	LMT-GP (Ours)	**4.262**	**0.628**	**4.509**	**0.491**
DICM	GLADNet [8]FG'18	5.152	0.396	3.400	0.265
	KinD [11] MM'19	4.826	0.560	3.150	0.436
	LLFormer [7]AAAI'23	**3.691**	0.585	**3.843**	0.468
	LLFlow [9] AAAI'22	4.189	**0.625**	**4.098**	**0.549**
	LMT-GP (Ours)	**3.682**	**0.631**	3.661	**0.591**
MEF	GLADNet [8]FG'18	5.294	0.406	3.578	0.222
	KinD [11] MM'19	5.066	0.521	3.463	0.324
	LLFormer [7]AAAI'23	**3.681**	0.624	4.448	**0.444**
	LLFlow [9] AAAI'22	4.531	**0.651**	**4.940**	0.428
	LMT-GP (Ours)	**3.899**	**0.673**	**5.078**	**0.503**

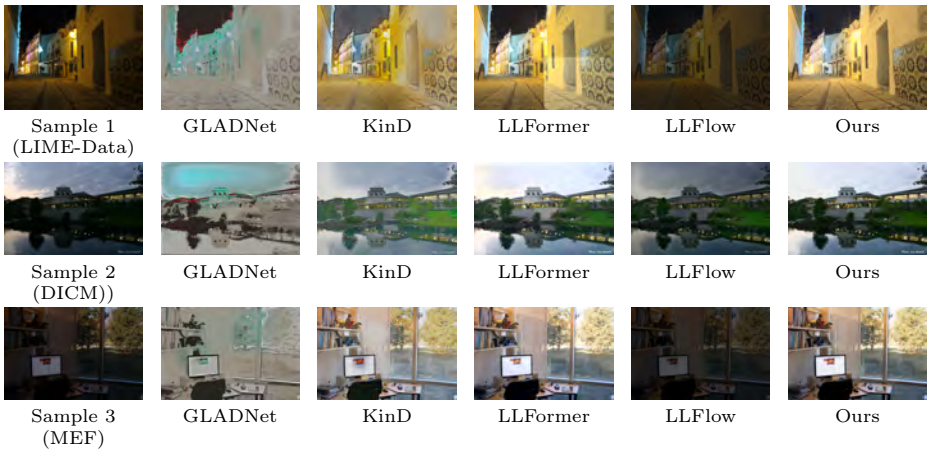


Fig. 2: Visual comparison of state-of-the-art LLIE methods on the unlabeled low-light datasets.

and outdoor natural lighting. The compared methods generate images with noise, color loss, artifacts, and insufficient brightness enhancement due to their limited denoising and generalization abilities. However, our method outperforms in terms of image quality due to our effective utilization of unlabeled data and latent vectors.

2.3 More Visual Results

This section is an extension of the results in Section 4.2 of the main paper. We demonstrate more visual comparison results on the LOL, VE-LOL, Darkface, and ACDC datasets. Meanwhile, we compare our method with 15 state-of-the-art LLIE methods.

As shown in Fig. 3 and Fig. 4, we select Sample 4 and Sample 5 from the LOL and VE-LOL datasets, respectively. We observe that the results generated by traditional method (LIME), unsupervised methods (EnGAN and PairLIE), and zero-shot methods (Zero-DCE++, RUAS, SCI, and RetinexDIP) exhibit under/over exposure due to their limited prior knowledge. For supervised methods such as RetinexNet, DeepUPE, and KinD, the generated images suffer from color distortion, artifacts, and the loss of texture due to the lack of color constraints during decomposing images based on the Retinex theory. Other supervised methods introduce noises in Sample 5 because of the limited denoising and generalization abilities, especially GLADNet. The DRBN method based on semi-supervised learning demonstrates strong generalization ability in Sample 5 but exhibits low saturation in Sample 4. Our method successfully suppresses noise and preserves rich color and texture details. Furthermore, our method achieves the highest PSNR and SSIM scores, indicating superior generalization performance.

To further validate the generalization performance of our method, we select more real-world images from Darkface and ACDC datasets, which are captured in different street scenarios. As shown in Fig. 5, our method accurately detects more objects and achieves the best visual effects. Furthermore, our method yields the high-quality segmentation results with less artifacts as shown in Fig. 6.

References

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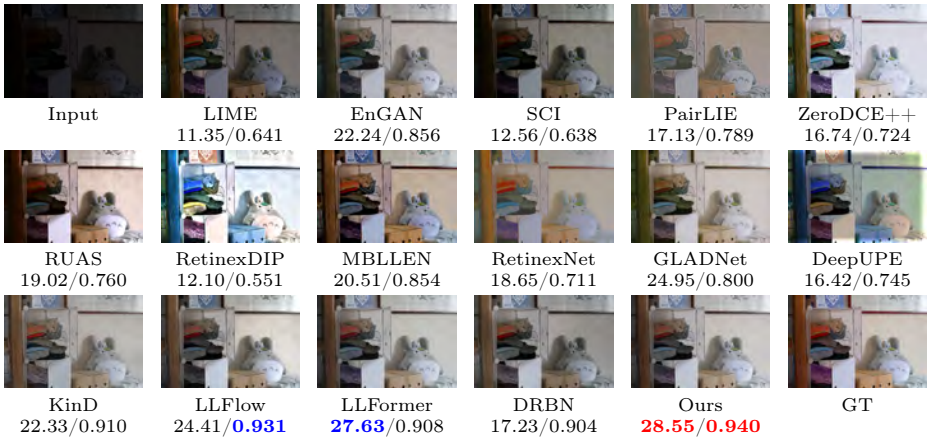


Fig. 3: Visual comparison of state-of-the-art LLIE methods on the LOL dataset (Sample 4). "*/*" represents PSNR/SSIM. The best result is in **red** whereas the second best one is in **blue**.

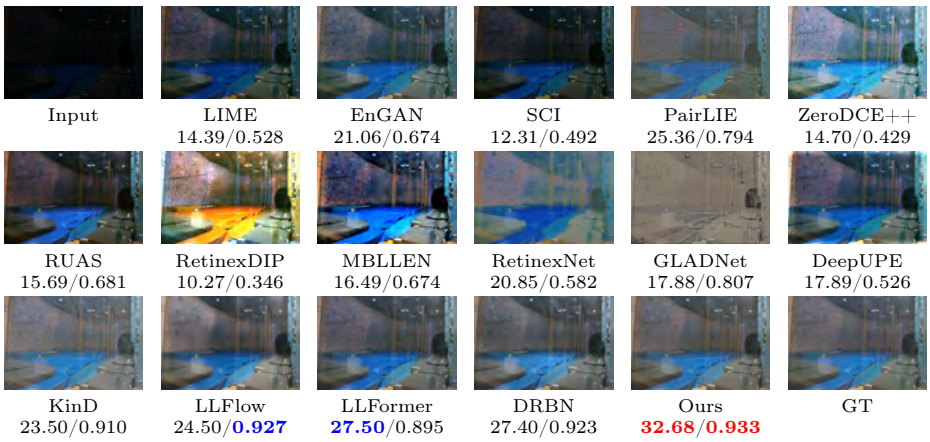


Fig. 4: Visual comparison of state-of-the-art LLIE methods on the VE-LOL dataset (Sample 5). "*/*" represents PSNR/SSIM. The best result is in **red** whereas the second best one is in **blue**.

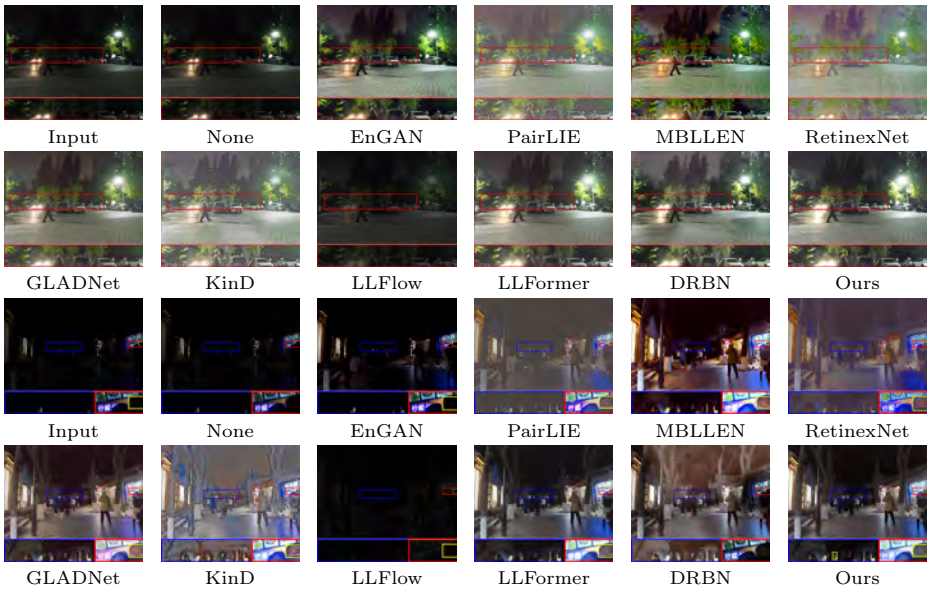


Fig. 5: More visual comparison results of detection on the Darkface dataset.

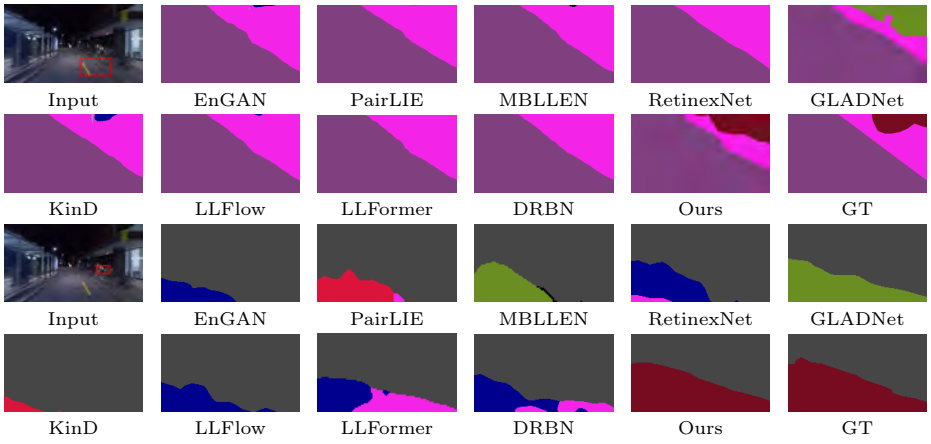


Fig. 6: More visual comparison results of semantic segmentation on the ACDC dataset.

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